

# AMBIKA SHARMA

+91-9327314062 ◊ Ahmedabad, Gujarat

ambika.pawan01@gmail.com ◊ linkedin.com/in/ambika-sharma-6a3517250 ◊ github.com/itsambi

## EDUCATION

### Vellore Institute of Technology

*Integrated M.Tech in Computer Science and Engineering (Cybersecurity)*

### Eklavya Public School, Ahmedabad, Gujarat

*Class XII*

### Eklavya Public School, Ahmedabad, Gujarat

*Class X*

Oct 2022 – Present

*GPA: 8.0*

Mar 2020 – Jun 2021

*Percentage: 84%*

Mar 2018 – Jun 2019

*Percentage: 82%*

## EXPERIENCE

### Cybersecurity Intern

*Skyislimit Technologies Pvt. Ltd.*

Nov 2025 – Dec 2025

*Mumbai, India*

- Analyzed 50+ web application requests and responses using Burp Suite to evaluate authentication workflows, session management mechanisms, and access control implementations.
- Conducted security assessments across 10+ application workflows, identifying authentication, authorization, and input validation weaknesses aligned with OWASP Top 10 principles.
- Produced 15+ structured security assessment reports documenting findings, risk observations, and remediation recommendations following industry reporting practices.

## PROJECTS

### Enhancing Malware Detection with T5 Generation and RoBERTa Attention Mechanisms

GitHub

*Python, PyTorch, Transformers, Scikit-learn, Google Colab*

- Built a transformer-based malware detection pipeline using the MalBehav-1 dataset containing 3,940 API call sequences; achieved 95.43% accuracy, 96.51% F1-score, and 97.53% AUC-ROC.
- Implemented a hybrid architecture combining T5-based next-API prediction with a customized RoBERTa model enhanced by an 8-head attention mechanism for behavioral malware classification.
- Processed 175 behavioral API features and implemented class-imbalance handling techniques across 3,940 samples to improve malware detection performance and model robustness.

### Automated Digital Forensics Investigation Toolkit

GitHub

*Python, SQLite, Pandas*

- Developed a Python-based DFIR toolkit integrating MD5, SHA1, and SHA256 hashing algorithms for automated evidence integrity verification and suspicious file identification.
- Built browser forensic modules capable of extracting and analyzing 1,000+ browsing history and download artifacts for user activity reconstruction and digital evidence collection.
- Designed a timeline reconstruction engine to correlate file system events, browser artifacts, and threat indicators, generating structured investigation reports for incident response workflows.

## SKILLS

**Security Tools:** Burp Suite, Nessus, Wireshark, Nmap, Ghidra, OWASP ZAP

**Machine Learning:** PyTorch, Scikit-learn, Hugging Face Transformers, RoBERTa, T5, Pandas, NumPy

**Programming & Dev Tools:** Python, SQL, HTML, CSS, GitHub, Google Colab

**Core Security Concepts:** Vulnerability Assessment, Web Application Security, Malware Analysis, Digital Forensics, Incident Response, Threat Intelligence

**Computer Science Fundamentals:** Operating Systems, Computer Networks, DBMS, OOPS

## EXTRA-CURRICULAR ACTIVITIES

- Student Co-ordinator: Coordinated logistics, volunteer management, and discipline activities for university events involving 100+ participants.
- Social Media & Event Lead: Managed promotional activities and supported the organization of 5+ technical workshops and student engagement events.
- National Service Scheme (NSS): Participated in community service initiatives focused on teamwork, social awareness, and public welfare activities.